

LSIS-AFS Competition — Submission Report

Replace all placeholder text in [brackets] with your team's information.
Delete any sections marked (optional) if not applicable. Submit this
file as SUBMISSION.md in the root of your repository.

Team Information

Field	Details
Team Name	[Your team name]
Members	[Name 1, Name 2, ...]
Institution / Affiliation	[University, company, or independent]
Contact Email	[Primary contact email]
Repository URL	[Link to your private repo]

Implementation Overview

Language & Technology Stack

Component	Choice
Primary Language	[e.g. C, Rust, Python, Go, Java]
Build System	[e.g. CMake, Cargo, Make, Gradle]
Key Libraries	[List any third-party libraries used]
Test Framework	[e.g. CTest, pytest, Go test, JUnit]
Platform(s) Tested	[e.g. Linux x86_64, macOS ARM64, Windows]

Architecture Summary

[2-3 paragraphs describing your system architecture, module structure, and key design decisions. Include a diagram if helpful.]

Build & Run Instructions

Prerequisites

[List all dependencies and how to install them.]

[package manager install commands, e.g.:]

[apt install ... / brew install ... / pip install ...]

Build

[exact build commands]

Run Tests

[exact test commands]

Run Examples

[exact commands to run key examples]

Gateway Status

Mark each gateway as Complete, Partial, or Not Attempted. Provide a brief note for partial implementations.

Gateway	Status	Notes
0 — Design & Architecture	☐ Complete / ☐ Partial / ☐ Not Attempted	
1 — Spreading Code Generation	☐ Complete / ☐ Partial / ☐ Not Attempted	
2 — Forward Error Correction	☐ Complete / ☐ Partial / ☐ Not Attempted	
3 — Navigation Message Framing	☐ Complete / ☐ Partial / ☐ Not Attempted	
4 — Baseband Signal Generation	☐ Complete / ☐ Partial / ☐ Not Attempted	
5 — Frame Sync & Decoding	☐ Complete / ☐ Partial / ☐ Not Attempted	
6 — Message Parsing	☐ Complete / ☐ Partial / ☐ Not Attempted	
7 — Integration & Validation	☐ Complete / ☐ Partial / ☐ Not Attempted	
8 — Documentation & Examples	☐ Complete / ☐ Partial / ☐ Not Attempted	

Validation Results

Gateway 1: Spreading Codes

Check	Pass/Fail	Evidence
Gold codes match Annex3 (all 210 PRNs) Weil primary codes match Annex3 (all 210 PRNs) Weil tertiary codes match Annex3 (all 210 PRNs) Secondary codes match Table 10 Tiered codes maintain coherency Code lengths match Table 9 Generation time < 1s per PRN		[test output or file reference]

Gateway 2: Forward Error Correction

Check	Pass/Fail	Evidence
BCH encoder produces valid 52-symbol codewords		
LDPC encoder produces valid codewords		
LDPC puncturing handled correctly		
CRC-24 matches specification		
Interleaver pattern validated		
Round-trip encode/decode recovers data		
Encoding time < 100ms per frame		
Decoding time < 1s per frame		
BER < 10^{-5} at SNR > 0 dB		

Gateway 3: Navigation Message Framing

Check	Pass/Fail	Evidence
Frame structure matches Figure 9		
Symbol counts: $68 + 52 + 5880 = 6000$		
Bit allocations match spec tables		
Frame duration is 12 seconds		

Check	Pass/Fail	Evidence
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Gateway 4: Baseband Signal Generation

Check	Pass/Fail	Evidence
I/Q samples correctly formatted		
AFS-I chip rate: 1.023 Mchip/s		
AFS-Q chip rate: 5.115 Mchip/s		
Symbol rate: 500 symbols/s (AFS-I)		
Code synchronization correct		
Signal duration: 12 seconds		

Gateway 5: Frame Sync & Decoding

Check	Pass/Fail	Evidence
Frame sync detection > 99% at SNR > 0 dB		
Decoders recover original data		
CRC validation catches errors		
LDPC converges in < 50 iterations		
Decode time < 1s per frame		

Gateway 6: Message Parsing

Check	Pass/Fail	Evidence
All subframes parse correctly		
WN, ITOW, TOI fields extracted		
Time of transmission accurate		
All SB3/SB4 message types handled		

Gateway 7: Integration

Check	Pass/Fail	Evidence
Round-trip recovers data with 100% accuracy		
All 12 interim test codes (Table 11) working		
Process 12s frames in < 1 second		
Interoperability demonstrated		
All “shall” requirements verified		

Performance Benchmarks

Metric	Your Result	Target
Code generation (per PRN)	[time]	< 1 second
Frame encoding (per frame)	[time]	< 100 ms
Frame decoding (per frame)	[time]	< 1 second
Real-time factor	[x real-time]	> 1x
BER at SNR 0 dB	[value]	< 10^{-5}
Frame sync reliability	[%]	> 99%
Test coverage	[%]	> 90%

[Describe your benchmarking methodology: hardware, OS, compiler flags, number of runs, etc.]

Interoperability (optional)

Partner Team	Test Performed	Result
[Team name]	[e.g. Decoded their signal file]	[Pass/Fail + notes]
[Team name]	[e.g. They decoded our signal file]	[Pass/Fail + notes]

[Describe the interop testing process and any issues encountered.]

Test Summary

Category	Total	Passing	Failing	Skipped
Unit tests				
Integration tests				
Compliance tests				
Performance tests				

[Paste or reference your test output here]

Known Limitations

[List any known issues, incomplete features, or deviations from the specification.]

Innovation & Extras (optional)

[Describe any additional features, optimizations, visualizations, or novel approaches your team implemented beyond the base requirements.]

File Manifest

List the key files and directories in your submission:

Path	Description
README.md	Project overview and quick start
SUBMISSION.md	This submission report
src/	Source code
tests/	Test suite
docs/	Documentation
examples/	Usage examples
[other]	[description]

Self-Assessment

Rate your submission against the scoring criteria:

Category (Points)	Self-Score	Justification
Correctness (40)	/40	
Performance (20)	/20	
Completeness (20)	/20	
Code Quality (10)	/10	
Innovation & Extras (10)	/10	
Total	/100	

Additional Notes (optional)

[Any other information you'd like the judges to know.]